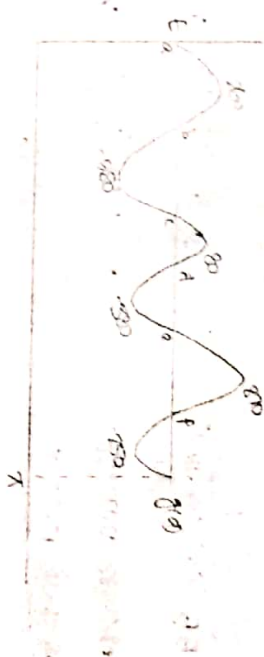


2) The living moment diagram for a petrol engine is drawn to a vertical scale of 1 mm - 500 Nm and a horizontal scale of 1 mm - 3°. The living moment dia. represents itself after every half turn of crankshaft. The areas above and below the mean torque line are 240 - 580 mm² - 380 mm² - 250 mm². Piston rods have a mass of 65 kg and the radius of gyration is 21 mm. If the engine speed is 1600 rpm. Determine the coeff. of fluctuations of speed.



Energy at $a = E$, $1 \text{ mm}^2 = 500 \text{ Nm} \times \frac{3 \times \pi}{180}$
 $b = E + 260$, $= F$

$c = E + 260 - 580$
 $= E - 320$

$d = E - 320 + 80$
 $= E - 240$

$e = E - 240 - 380$
 $= E - 620$

$f = E - 620 + 810$
 $= E - 250$

$$g = E + 250 - 250$$

• E

$$E_{\text{man}} = (1 + 250) - (E - 620)$$

$$= 880 \text{ units}$$

$$= 880 \times 500 \times \frac{3}{160} \times 1 \text{ Nm} = 23038.4 \text{ Nm}$$

$$K_s = \frac{E_{\text{man}}}{I \omega^2}$$

$$= \frac{23038.4}{55 \times (2.1)^2 \times \left(\frac{2\pi \times 1600}{60}\right)^2}$$

$$= 0.00338$$

$$= 0.33\%$$

$$E_{\text{man}} = \frac{1}{2} I \omega_1^2 - \frac{1}{2} I \omega_2^2$$

$$= \frac{1}{2} I (\omega_1^2 - \omega_2^2)$$

$$= \frac{1}{2} I (\omega_1 + \omega_2) (\omega_1 - \omega_2)$$

$$= \frac{1}{2} I \left(\frac{\omega_1 + \omega_2}{2} \right) (\omega_1 - \omega_2)$$

$$= I \omega \left(\frac{\omega_1 - \omega_2}{2} \right) \omega$$

$$= I \omega^2 K_s$$

$$K_s = \left(\frac{\omega_1 - \omega_2}{2 \omega} \right)$$

For the above loading moment the speed should be the sum of flywheel if the speed should be maintained within the limit $1600 \pm 50 \text{ rpm}$.

$$K_s = \frac{100}{1600}$$

$$I = \frac{E_{\text{man}}}{K_s \omega^2} = \frac{23038.4}{0.0625 \times \left(\frac{2\pi \times 1600}{60}\right)^2} = 13.13 \text{ kgm}^2$$

$$\omega_1 = 1650 \text{ rpm}$$

$$\omega_2 = 1550 \text{ rpm}$$

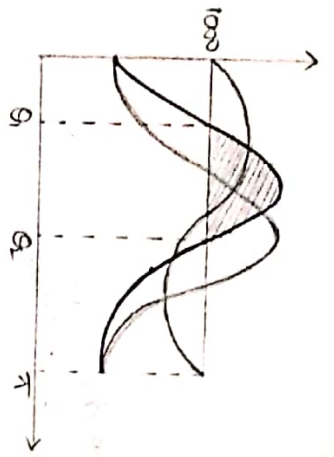
$$K = \frac{\omega_1 - \omega_2}{\omega}$$

$$= \left(\frac{2\pi \times 1650}{60} - \frac{2\pi \times 1550}{60} \right) = \frac{100}{1600}$$

1) The torque delivered by a 2 stroke engine is represented by $T = 1000 + 300 \sin 70 - 500 \cos 20$ where θ is the angle turned by the crank from the inner dead centre. The engine speed is 250 rpm. Mass of the flywheel is 400 kg and radius of gyration is 400 mm. Determine

- Power developed
- Total % fluctuation of speed
- Ang. acc'n of flywheel when the crank has rotated to any angle of 60° from the IDC
- Min ang acc'n and retardation of the flywheel.

$$T = (1000 + 300 \sin 2\theta - 500 \cos 2\theta) \text{ Nm}$$



$$i) P = T_{mean} \times \omega = 1000 \times \frac{2\pi \times 250}{60}$$

$$= 26179.9 \approx 26180 \text{ W}$$

$$T_{mean} = \frac{\int_0^\pi T d\theta}{\pi}$$

$$T = T_{mean}$$

$$\int_0^\pi T d\theta = E_{mean}$$

$$K_s = \frac{E_{mean}}{I \omega^2}$$

$$T = T_{mean}$$

$$1000 + 300 \sin 2\theta - 500 \cos 2\theta = 1000$$

$$300 \sin 2\theta = 500 \cos 2\theta$$

$$\tan 2\theta = \frac{5}{3}$$

$$2\theta = 59.03^\circ, 239.03^\circ$$

$$\theta = 29.51^\circ, 119.51^\circ$$

$$E_{mean} = \int_{29.51}^{119.51} T d\theta = E_{mean}$$

$$E_{mean} = \int_{29.51}^{119.51} (300 \sin 2\theta - 500 \cos 2\theta) d\theta$$

$$= (-150 \cos 2\theta - 250 \sin 2\theta) \Big|_{29.51}^{119.51}$$

$$= 583.09$$

$$K_s = \frac{E_{mean}}{I \omega^2} = \frac{583.1}{400 \times (6.4)^2 \times \left(\frac{2\pi \times 250}{60}\right)^2}$$

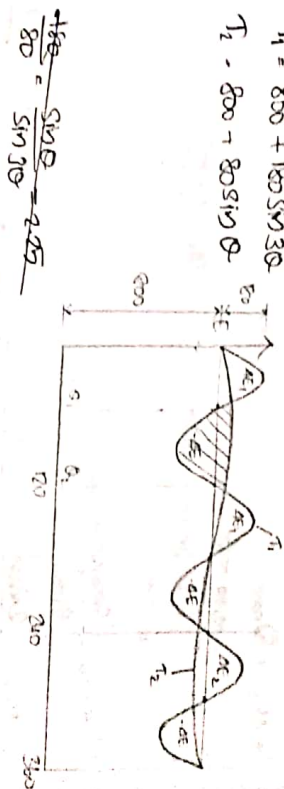
$$= 0.01329 = 1.329$$

- 2) A machine is coupled to a 2 stroke engine which produces a torque of $800 + 180 \sin 3\theta$ Nm. where θ is the crank angle. The mean engine speed is 500 rpm. The flywheel and the other rotating parts attached to the engine have a mass of 350 kg at a radius of gyration 220 mm. Calculate the power of the engine and the total fluctuation of speed of the flywheel. ~~there is still torque is~~ existing

$$T_1 = 800 + 180 \sin 3\theta$$

4)

$$T_2 = 800 + 80 \sin \theta$$



$$\frac{T_1}{80} = \frac{\sin \theta}{\sin 3\theta} = 2.25$$

$$\frac{45 \sin 3\theta}{80} = \frac{\sin \theta}{\sin 3\theta} \approx 2.25$$

$$\frac{35 \sin \theta}{80} = \frac{\sin \theta}{\sin 3\theta} \approx 2.25$$

$$\sin 3\theta = 0.25 \quad (3 \sin \theta - 4 \sin^3 \theta)$$

$$800 + 180 \sin 30^\circ = 800 + 80 \sin \theta \quad \leftarrow (\pi, \pi)$$

$$180 (3 \sin \theta - 4 \sin^3 \theta) = 80 \sin \theta$$

$$180 \sin \theta (3 - 4 \sin^2 \theta) = 80 \sin \theta$$

$$\sin^2 \theta = 0.637$$

$$\sin \theta = \pm 0.799$$

$$0, 52.92^\circ, 127.02^\circ, 180^\circ, 232.92^\circ, 307.02^\circ$$

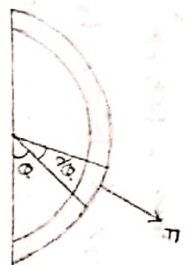
e - part

$$+ 4E_2 = Q_{max}$$

$$e = \int_{0.5142}^{0.5142} (180 \sin 30^\circ - 80 \sin \theta) d\theta = 208.3 \text{ Nm}$$

$$K = \frac{Q_{max}}{I_{xx}}$$

$$= \frac{208.3}{350 \times 0.2^2 \times \left(\frac{20 \times 400}{60}\right)^2} = 0.007 = 0.7\%$$



$$\int_0^\pi P(R \sin \theta) R \omega^2 \sin \theta$$

$$F = m R \omega^2$$

$$= (P R \sin \theta) R \omega^2$$

$$\text{vertical force} = \int_0^\pi (G F \sin \theta)$$

$$= \int_0^\pi P R \sin \theta R \omega^2 \sin \theta$$

$$= P R^2 \omega^2 \int_0^\pi \sin^2 \theta d\theta$$

$$= [\cos \theta]_0^\pi P R^2 \omega^2$$

$$= (1 - 1) P R^2 \omega^2$$

$$= P R^2 \omega^2 \times 2$$

$$2 P R^2 \omega^2 = \sigma A L$$

$$\sigma = P R^2 \omega^2$$

$$= \int V^2$$

2) The trailing moment dia of multicylinder engine is
 diameter is a vertical scale of mm = 650 mm. Horizontal
 scale of mm = 4.5°. Areas above & below the mean
 torque line are -2, +380, -260, +310, -300, +242, -380,
 +265 and -229 mm. The fluctuations of speed is limited
 to $\pm 1.8\%$ of mean speed which is 600 rpm. The
 density of the material of the ring is 700 kg/m^3 and
 width of ring is 4.5 times its thickness. The centrifugal
 stress in ring material is limited to 6 N/mm^2 . If the
 loss of the arms is negligible 5% of 1102 determine
 the dia & v.s of Flywheel

$$L_{\text{mean}} = 22360 \text{ Nmm}$$

$$\text{Fluctuation} = \pm 0.018$$

$$K_s = 0.036$$

$$\sigma = 6 \text{ N/mm}^2 = 6 \times 10^6 \text{ N/m}^2$$

$$20000 \times 1000 \text{ kg/m}^3 \times v^2 (\text{m/s})^2$$

$$v = 29 \text{ m/s}$$

$$v = r\omega$$

$$r = v/\omega = \frac{29}{(2\pi \times 600)} = 0.1 \text{ m}$$

$$I = \frac{L_{\text{mean}}}{K_s \omega^2}$$

$$= \frac{22360}{0.036 \times \left(\frac{2\pi \times 600}{60}\right)^2} = 354 \text{ kg/m}^2$$

mean fluctuation
 mean fluctuation
 fluctuation = 1.8%
 $K_s = 0.036$

$$I_{\text{rim}} = 0.95 \times 354$$

$$= 336.3 \text{ kgm}^2$$

$$R = 0.1 \text{ m}$$

$$I_{\text{arm}} = m_{\text{arm}} R^2$$

$$m_{\text{arm}} = \frac{I_{\text{arm}}}{R^2} = \frac{336.3}{0.019}$$

$$= 686.32 \text{ kg}$$

$$m_{\text{arm}} = 2\pi R A \rho$$

$$A = \frac{m_{\text{arm}}}{2\pi R \rho} = \frac{686.32}{2\pi \times 0.1 \times 7000} = \frac{686.32}{2\pi \times 700} = 0.22$$

$$= 11 = 4.15 \text{ t}^2$$

$$t = 0.01 \text{ m}$$

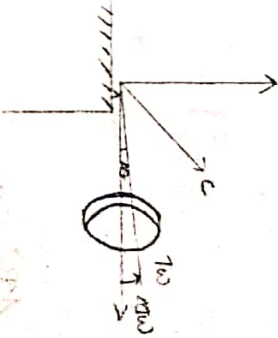
Cyclocope

$$\frac{d(I_{\text{rim}})}{dt} = \frac{d(I_{\text{rim}})}{dt}$$

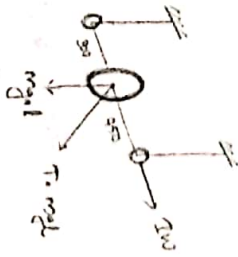
$$= \frac{d}{dt} \left(\frac{I_{\text{rim}}}{dt} \right)$$

$$= I_{\text{rim}} \times \frac{d\theta}{dt}$$

$$I_{\text{rim}} \times \frac{d\theta}{dt} = C = \text{mg}$$



1) Disc supported by 2 bearings on a shaft of negligible weight has a mass of 50 kg and a radius of gyration 500 mm. The distance of the disc from the bearings are 800 mm to the right from left hand and 400 mm to the left from right hand bearing. The bearing on supported by two vertical rods. When the disc rotates at 100 rad/s in the clockwise direction looking from the left hand bearing the base supporting the left hand side bearing breaks. Find the ang vel. of this section at the instant the disc is out and discuss the motion of disc.

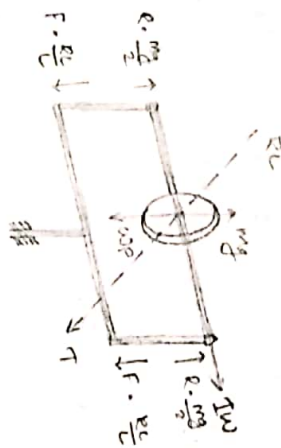


$$\frac{4T\omega}{At} = T\omega$$

$$C - T\omega$$

$$\omega_p \cdot \frac{c}{T\omega} = \frac{mgl}{T\omega} = \frac{80 \times 9.8 \times 0.45}{80 \times 0.3 \times 100} = 0.49 \text{ rad/s}$$

2) A disc with $k = 60 \text{ mm}$ and a mass of 4 kg is mounted centrally on a horizontal and of 800 mm length with the bearings. It spins about the axis at 800 rpm and when viewed from the right hand side bearing. The axis is vertical and at 50 mm in the horizontal direction when viewed from above. Determine the angular motion of each bearing due to wear and gyroscopic effects.



$$R = \frac{m\omega}{2} \cdot \frac{4 \times 9.8}{2} = 19.6$$

$$R_c = T\omega$$

$$F = \frac{R_c}{c} = \frac{4 \times 0.06 \times \frac{2\pi \times 800}{60} \times \frac{2\pi \times 50}{60}}{60}$$

3) An airplane flying at 200 km/hr turns towards the left and completes a quarter circle of 60 m radius. Mass of the rotating engine and propeller of plane is 500 kg with $k = 500 \text{ mm}$. The engine speed is 2000 rpm when it is turned from the 200, determine the gyroscopic couple on the aircraft and state its effects

112. m/s

$$= 450 \times 6320^\circ$$

$$= 96.4 \text{ kg m}^2$$

$$\omega = 2000 \text{ rpm} = 2094 \text{ rad/s}$$

$$V = 200 \text{ m/s}$$

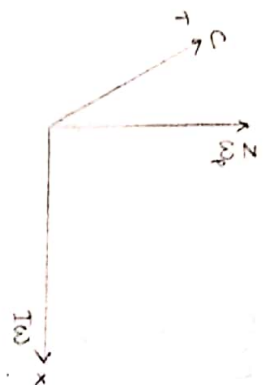
$$\omega_p = \frac{200 \times \pi}{60}$$

$$= 1.05 \text{ rad/s}$$

$$C = I \omega_p = 46.5 \times 1.05 = 48.825 \text{ Nm}$$

$$= 10710 \text{ Nm}$$

If the propeller is turning the aircraft's heading from 200 and aircraft is taking a left turn what will be the gyroscopic effects.



Thrust - engine
Turning - Propeller
Speed - rpm

1) turbine engine of a ship has a mass of 2.2 tons and rotates at 1800 rpm. It is when both from the aft. Radius of gyration of the engine is 320 mm. Determine the gyroscopic couple and its effects when

i) the ship's trim is right at a radius 250 m with a speed of 25 km/hr.

ii) the ship's pitching with the bow rising at an ang vel. of 0.8 rad/s

1. m/s

$$= 2.2 \times 10^3 \times (320 \times 10^{-3})^2$$

$$= 225.28 \text{ kg m}^2$$

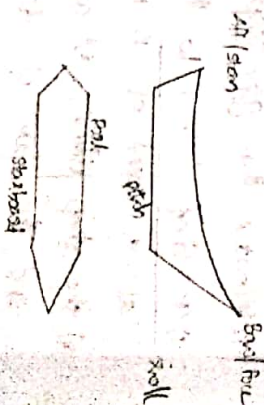
$$\omega = 1800 \text{ rpm}$$

$$= 188.5 \text{ rad/s}$$

$$\omega_p = \frac{25 \times \pi}{18}$$

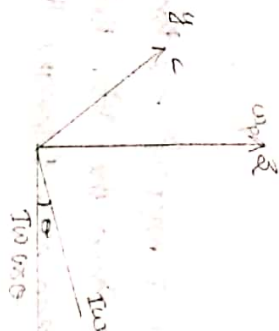
$$= 250$$

$$= 0.28$$



$$I_{\text{roll}} = 215.28 + 188.5 + 0.028$$

$$= 1184.02 \text{ Nms}$$



C - I_{roll}

$$\omega_p = \frac{C}{(I_{roll})} = \frac{C}{I_{roll}}$$

2) The rotor of the turbine of a ship has a mass of

2500 kg and rotates at a speed of 3200 rpm (rev

when viewed from the stern/aft. The rotor has $k = 0.4 \text{ m}$

determine the gyroscopic couple and its effect when

i) the ship turns to the left in a curve of 80 m radius

at a speed of 15 knots (1 knot = 1852 m/hr)

ii) the ship pitches 5° above and 5° below the normal position

and the bar is descending with its own velocity

The pitching motion is simple harmonic with a period

time of 40 s.

iii) the ship rolls at the instant with ang. vel. 0.4 rad/s and

when viewed from the stern. Also find the max ang. vel. during pitching

$$1) \omega_p = \frac{V}{R}$$

$$ii) V_{max} = A \omega$$

$$A = 5 \cdot 5 \cdot \frac{\pi}{180}$$

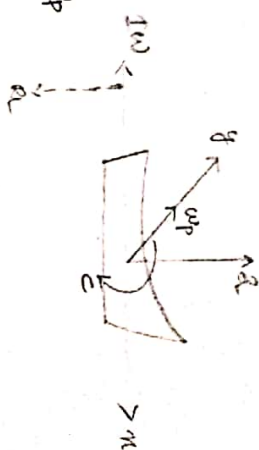
$$\theta = 2\pi$$

$$t = 40$$

$$\frac{2\pi}{40} \cdot \omega$$

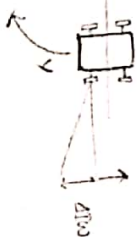


Reaction couple = I_{roll} ω_{roll}



$$= 2500 \times 0.4 \times \frac{2\pi \times 3200}{60} \times 0.013$$

$$= 1742.53$$

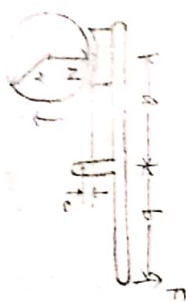


$$\frac{m v^2}{R}$$

Brakes

1. shoe brake (block brake)
2. Band brake
3. Band & Block "
4. Internal expanding "

Block brake



~~For N.A.C~~

$$F \cdot b - N \cdot a - \mu N \cdot c = 0$$

$$F_b = \mu N r$$

$$N = \frac{F_b}{\mu}$$

$$F = \frac{N(a + \mu c)}{b}$$

C.C.V

$$F \cdot b - N \cdot a + \mu N \cdot c = 0$$

$$F = \frac{N(a - \mu c)}{b}$$

Q. A block brake has shown in fig has a brake drum dia of 1 m. the brake has to provide a braking torque of 2110 N.m at 400 rpm. the width of friction is 0.32 m. dimensions are $a = 150 \text{ mm}$, $b = 800 \text{ mm}$, $c = 25 \text{ mm}$. find the force required to apply the brake for cw & ccw rotation.

Q2

$$F = \frac{N(a+M)}{b}$$

$$N = \frac{T_b}{\mu} = \frac{240}{0.32 \times 0.5} = 1500$$

$$F = \frac{1500 (0.15 + 0.32 \times 0.05)}{0.8} = 296.25 \text{ N}$$

Q3

$$F = \frac{N(a-M)}{b}$$

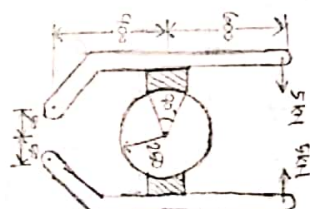
$$= \frac{1500 (0.15 - 0.32 \times 0.05)}{0.8} = 266.25 \text{ N}$$

If the angle subtended by the brake shoe at the center is 100°
Find the force required to actuate it

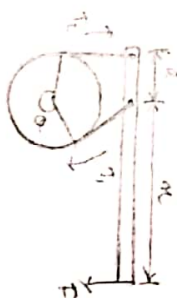
$$F_r = \mu N$$

$$M = \mu \left[\frac{9.81 \sin(90^\circ)}{0 + \sin 0} \right]$$

$$= 0.32 \left[\frac{4 \cdot \sin(90^\circ)}{\frac{100\pi \sin 100}{180}} \right] = 1.12 \times 0.35$$



Band brake



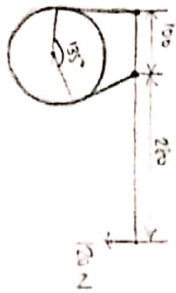
$$F_r = T_2$$

$$F = \frac{T_2 a}{r}$$

$$\frac{T_1}{T_2} = e^{\mu \theta}$$

$$(T_1 - T_2) \sim \mu \theta$$

2) A simple band brake is applied to a shaft carrying a flywheel of 250 kg mass and $k = 300 \text{ mm}$. The shaft speed is 200 rpm. Drum dia is 200 mm, $\mu = 0.25$. Determine the brake torque, when the force of 120 N is applied at the end of the lever. Also find No. of Turns of the flywheel and time taken before it comes to rest for two rotations.



$$T_2 = \frac{120 \times 0.28}{0.1} = \underline{\underline{336 \text{ N}}}$$

$$T_1 = T_2 \cdot e^{0.25 \times \frac{2\pi \times 1}{4}} = 6.64 \underline{\underline{596.82 \text{ N/m}}}$$

$$T_0 = T_2 (e^{2\theta} - 1) \cdot r$$

$$= \frac{F \cdot l}{a} (e^{2\theta} - 1) \cdot r$$

$$T_1 = \frac{F \cdot l}{a} \quad T_2 = \frac{T_1}{e^{2\theta}}$$

$$T_0 = (T_1 - T_2) \cdot r$$

$$= T_1 \left(1 - \frac{1}{e^{2\theta}}\right) \cdot r = \frac{F \cdot l}{a} \left(1 - \frac{1}{e^{2\theta}}\right) \cdot r$$

C.I.D

$$T_2 = 336 \text{ N}$$

$$e^{2\theta} = 2.66$$

$$T_1 = 336 \times 2.66 = 896 \text{ N}$$

$$T_0 = (896 - 336) \times 0.1 = \underline{\underline{56 \text{ N/m}}}$$

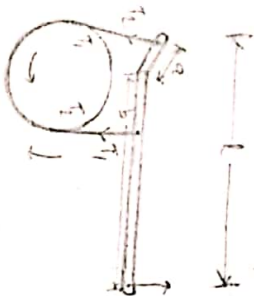
$$T = 21 \text{ N/m}$$

$$T = T_0 \Rightarrow \alpha = \frac{21}{250 \times 31} = 0.93$$

$$\alpha = \frac{(n_1 - n_2)}{t} \Rightarrow 0.93 = \frac{2\pi \times 200}{t} \Rightarrow t = \underline{\underline{22.5}}$$

$$\frac{1}{2} T_0 t = T(e)$$

Differential band brake



$$F \cdot l - T_b + T_a = 0$$

$$F = \frac{T_b - T_a}{l}$$

$$F \cdot l - T_b + T_a = 0$$

$$F = \frac{T_b - T_a}{l}$$

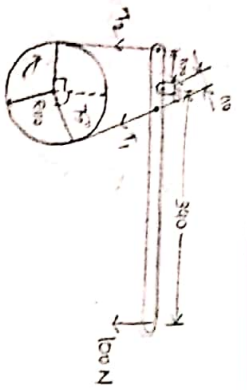
C.I.D

$$T_1 = 336 \text{ N}$$

$$T_2 = 336 \times 2.66 = 896 \text{ N}$$

$$T_0 = 21 \text{ N/m}$$

2)



A hand brake is shown is applied to a shaft carrying a flywheel of 300 kg mass with $k = 400 \text{ mm}$. Running at 340 rpm. Find the torque applied due to a pull of 100 N. If $\mu = 0.25$ find the no. of rev of flywheel before it comes to rest.

$$100 = 0.34 + T_1 \cdot 0.04 - T_2 \cdot 0.1 = 0$$

$$\frac{T_1}{T_2} = e^{(0.25 \cdot 175 \cdot \frac{\pi}{2})} = 2.34$$

$$T_2 = (T_1 - T_2) \cdot r$$

$$T_1 = 2.34 T_2$$

$$100 = 0.34 + 2.34 \cdot 0.04 T_2 - T_2 \cdot 0.1 = 0$$

$$54 = 6.4 \cdot 10^{-3} T_2$$

$$T_2 = 5312.5$$

$$T_1 = 2.34 \cdot 5312.5 = 12431.25$$

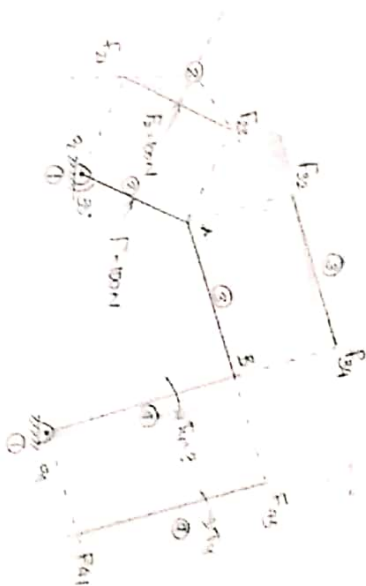
2) A diff hand brake has a dia of 800 mm. The 2 ends of

the band are fixed to the pins on the opposite of the fulcrum of the lever at distances of 40 mm and 200 mm from the fulcrum. Angle of contact is 270° and $\mu = 0.2$. Determine the maximum braking torque that can be obtained when a force of 800 N is applied at a distance of 800 mm from the fulcrum.



Force Analysis

Static Force Analysis



Link 2

→ 3 Forces (F_{12}, F_{23}, F_2)

3 forces will be in eqn only if they are concurrent, and they form a closed triangle.

Link 3

→ 2 forces (F_{32}, F_{34})

2 forces can be in eqn only if they are equal, opposite and collinear.

Link 4

→ 2 forces (F_{41}, F_{43}) & a torque T_4

2 forces must be equal & opposite / they should form a couple equal & opposite to T_4

2)

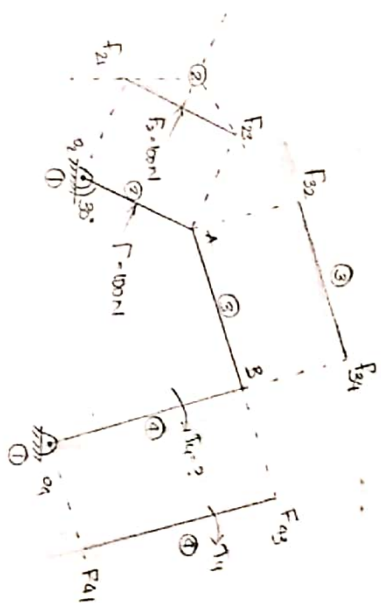


3)



Force Analysis

Static Force Analysis



Link 2

→ 3 forces (F_2, F_3, F_1)

3 forces will be in eqn only if they are concurrent, and they form a closed Δ^{ve}.

Link 3

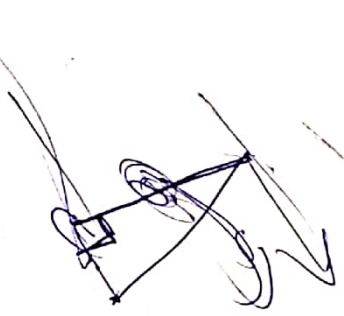
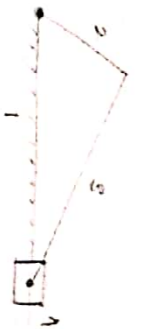
→ 2 forces (F_{32}, F_{34})

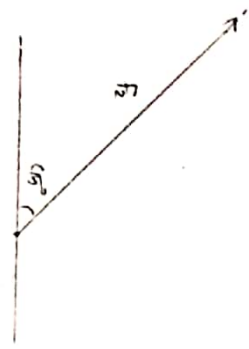
2 forces can be in eqn only if they are equal, opposite and collinear.

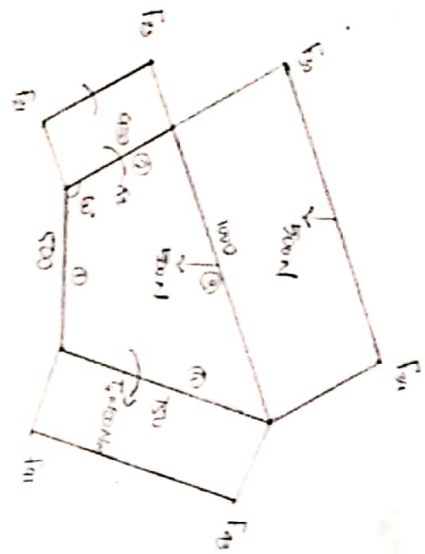
Link 4

→ 2 forces (F_{41}, F_{43}) & a torque T_1

2 forces must be equal & opposite / they should form a couple equal & opposite to T_1







Link 2

2 revs & 1 rev.

Link 3

3 revs.

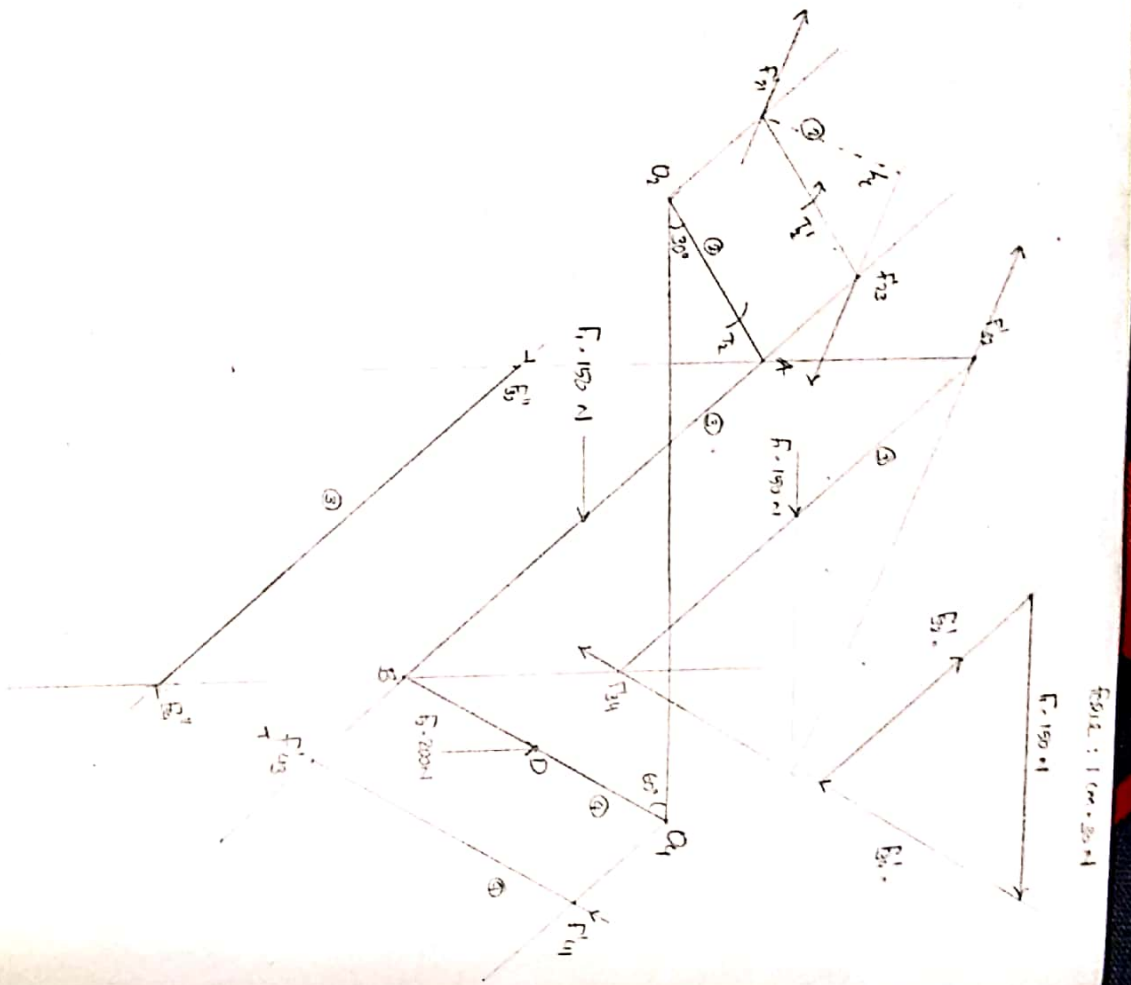
Link 4

2 revs.

From the link 3 ;

Scale 1 cm = 100 N





i) considering only F_1 is acting

$$F_1' = F_1' \times h_1$$

$$= 12.50 \text{ Nm}$$

ii) considering only F_2 is acting
scale $1 \text{ cm} = 10 \text{ N}$

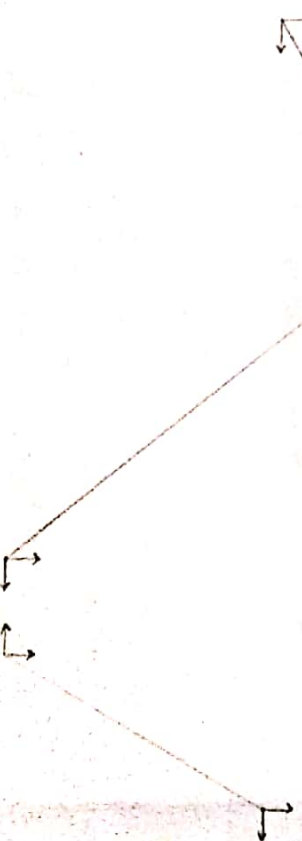
Force polygon by links ④

$$F_2'' = F_2'' \times h_2$$

$$= 50 \times 15 = 7.5 \text{ Nm}$$

$$= 12.5 - 7.5 = 5 \text{ Nm}$$

Alternative method



Assignment I

Link 2

$$\sum F_x = 0$$

$$F_{2m} + F_{2n} = 0$$

$$\sum F_y = 0$$

$$F_{2y} + F_{2z} = 0$$

$$\sum M = 0$$

$$I_2 - F_{2m} l_2 \sin \alpha_2 + F_{2y} l_2 \cos \alpha_2 = 0$$

Link 3

$$-F_{3m} + F_{3n} + F_{3y} = 0$$

$$F_{3y} - F_{3z} = 0$$

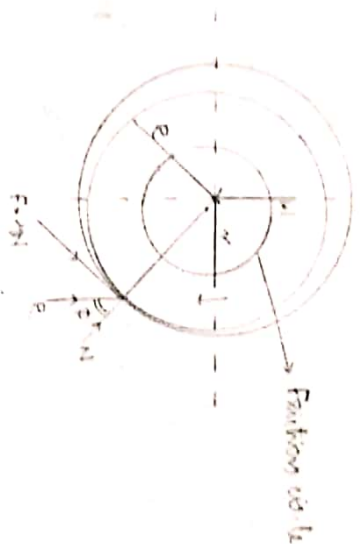
$$F_{3m} l_3 \sin \alpha_3 + F_{3y} l_3 \cos \alpha_3 - 150 \frac{l_3}{2} \sin \alpha_3 = 0$$

Link 4

$$F_{4m} - F_{4n} = 0$$

$$F_{4y} - F_{4z} + 200 = 0$$

$$F_{4y} l_4 \cos \alpha_4 - F_{4n} l_4 \sin \alpha_4 + 200 \frac{l_4}{2} \cos \alpha_4 = 0$$



Radius of friction circle

$$r = R \sin \phi$$

For small angle ϕ , $\sin \phi \approx \tan \phi$

$$r = \mu R$$

Forward Bearing :- Bearing without ball, roller, needle etc..

Forward is the position of shaft going inside the bearing.

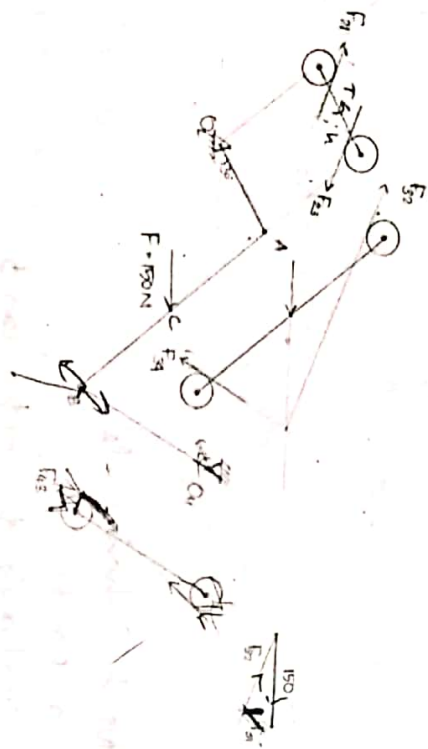
* The resultant force in the joint should be tangential to the friction circle whose radius = μR

* The reaction R (force on joint) produces a torque opposing to the direction of relative motion.

$$0.5 \times 5 = 2.5 \text{ m}$$

$$\left. \begin{array}{l} Q_A = 75 \text{ m} \\ B_C = AE/2 \\ Q_A = 15 \text{ m} \end{array} \right\} \text{1000 of journal } 5 \text{ m}$$

$$M = 0.5$$



2) What is the turning moment available to the crank considering the entire effects of pistons and connecting rod

$$m_p = 5 \text{ kg}$$

$$m_1 = 20 \text{ kg}$$

$$l_g = 100 \text{ mm}$$

$$\cos \theta = \frac{0.5}{1.3}$$

Velocity

$$V_A = V_G + V_{A/G}$$

$$0 = w_2 \cdot 0.1 - 10 \cdot 1000 - 10000 \text{ mm/s}$$

$$V_B = V_A + V_{B/A}$$

B is a point on the slider B shown
hence along the line of stroke.

$$V_{B/A} = 100 \text{ m/s}$$

$$w_3 \cdot AB = 100$$

$$w_3 = \frac{100}{35} = 2.857 \text{ rad/s} \curvearrowright$$

Acc'n

$$a_A = a_G + a_{A/G}$$

$$1000 \text{ mm/s}^2 = w_2^2 \cdot 0.1$$

$$a_3 = \frac{500}{35} = 14.28 \text{ rad/s}^2$$

$$a_B = 500 \text{ mm/s}^2$$

$$a_B = a_A + a_{B/A}$$

$$a_{B/A} = a_{B/A}^t$$

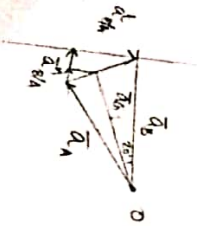
$$V_{B/A} = 100 \text{ m/s}$$

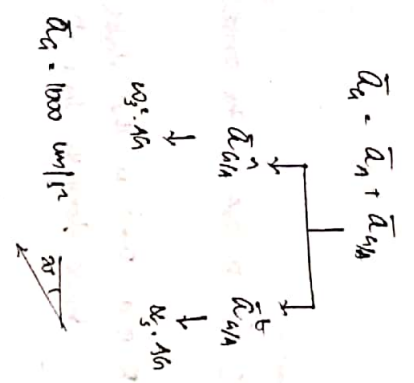
$$V_A = 100 \text{ m/s}$$



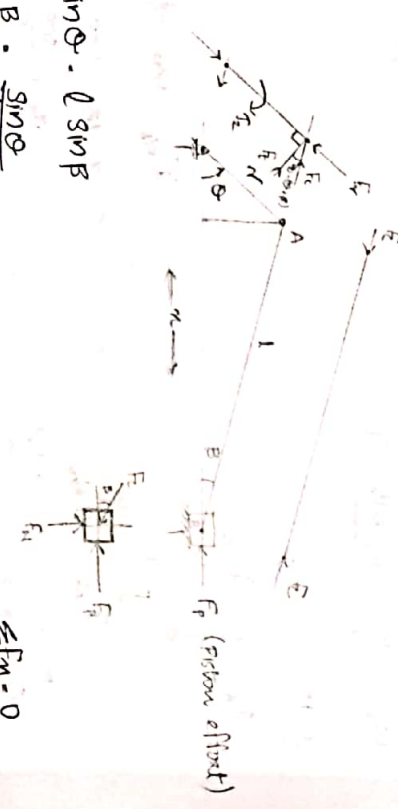
$$V_{G/A} = 90 \text{ m/s}$$

$$a_{B/A} = 500 \text{ mm/s}^2$$





Analytical method of force analysis in slider-crank mechanism



$$\begin{aligned}
 n \sin \theta &= l \sin \phi \\
 \sin \phi &= \frac{\sin \theta}{(l/r)} \\
 &= \frac{\sin \theta}{n} \\
 \cos \phi &= \sqrt{1 - \sin^2 \phi} \\
 &= \sqrt{1 - \frac{\sin^2 \theta}{n^2}} = \frac{\sqrt{n^2 - \sin^2 \theta}}{n} \\
 \frac{\sum F_x = 0}{F_c \cos \phi - F_p} &= 0 \\
 F_c &= \frac{F_p}{\cos \phi} \\
 \frac{\sum F_y = 0}{F_n - F_c \sin \phi} &= 0 \\
 F_n &= F_c \sin \phi
 \end{aligned}$$

$$\begin{aligned}
 F_n &= F_c \cos \phi \\
 F_n &= F_c \sin \phi \\
 &= F_p \tan \phi
 \end{aligned}$$

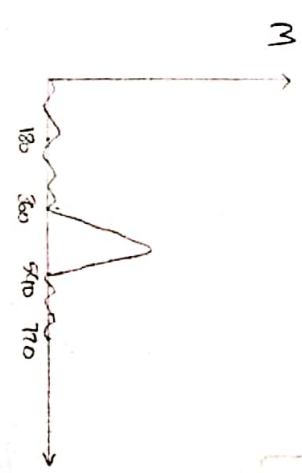
$$\begin{aligned}
 F_c &= F_c \sin(\theta + \phi) \\
 F_n &= F_c \cos(\theta + \phi)
 \end{aligned}$$

Turning moment available on the crankshaft

$$\begin{aligned}
 M &= F_c r \\
 &= F_c \sin(\theta + \phi) \cdot r \\
 &= \frac{F_p}{\cos \phi} \sin(\theta + \phi) \cdot r
 \end{aligned}$$

$$M = \frac{n r F_p}{\sqrt{n^2 - \sin^2 \theta}} \sin \theta \frac{\sqrt{n^2 - \sin^2 \theta}}{n} + \cos \theta \frac{\sin \theta}{n}$$

$$F_p = F_g + F_i$$



$$r = (r+l) - (r \cos \theta + l \cos \phi)$$

$$= (r+l) - (r \cos \theta + \frac{l}{n} \sqrt{n^2 - \sin^2 \theta})$$

$$= (r+l) - (r \cos \theta + \sqrt{n^2 - \sin^2 \theta})$$

$$\dot{r} = \frac{dr}{dt}$$

$$= -r \left[-\sin \theta \frac{d\theta}{dt} + \frac{1 \times -2 \sin \theta \cos \theta}{2 \sqrt{n^2 - \sin^2 \theta}} \frac{d\theta}{dt} \right]$$

$$= r \omega \left[\sin \theta + \frac{\sin^2 \theta}{\sqrt{n^2 - \sin^2 \theta}} \right]$$

Assume n is large.

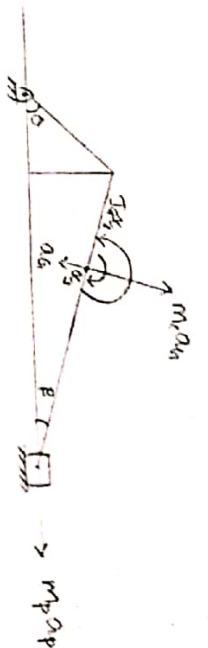
$$\dot{r} = r \omega \left[\sin \theta + \frac{\sin^2 \theta}{2n} \right]$$

$$\dot{r} = r \omega \left[\cos \theta \frac{d\theta}{dt} + \frac{2 \cos^2 \theta}{2n} \frac{d\theta}{dt} \right]$$

$$= r \omega^2 \left[\cos \theta + \frac{\cos^2 \theta}{n} \right]$$

$$M_p = m \cos \theta + r \sqrt{n^2 - \sin^2 \theta}$$

$$= m \cos \theta + l$$



$$\sin \phi = \frac{\sin \theta}{n}$$

$$\cos \phi \frac{d\phi}{dt} = \frac{\cos \theta}{n} \cdot \frac{d\theta}{dt}$$

$$M_c = \frac{m \cos \theta}{\cos \phi}$$

$$\frac{M_c}{n} = \frac{\cos \theta}{\sqrt{n^2 - \sin^2 \theta}}$$

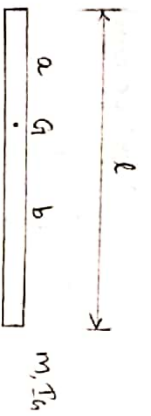
$$M_c = \frac{m \cos \theta}{\sqrt{n^2 - \sin^2 \theta}}$$

$$M_c = \frac{dM}{dt} \cdot \omega$$

$$\left[\frac{\sqrt{n^2 - \sin^2 \theta} \cdot (-\sin \theta) \frac{d\theta}{dt} - \cos \theta \cdot \frac{2 \sin \theta \cos \theta}{2 \sqrt{n^2 - \sin^2 \theta}} \frac{d\theta}{dt}}{n^2 - \sin^2 \theta} \right]$$

$$= \omega^2 \left[\frac{(n^2 - \sin^2 \theta) (-\sin \theta) + \sin \theta \cos^2 \theta}{(n^2 - \sin^2 \theta)^{3/2}} \right]$$

$$= \frac{\omega^2 \sin^3 \theta (1 - n^2)}{(n^2 - \sin^2 \theta)^{3/2}}$$



mass should be same

$$m_a + m_b = m$$

location of CG should be same

$$m_a \cdot a = m_b \cdot b$$

Same moment of inertia

$$m_a a^2 + m_b b^2 = I_G$$

$$m_a + m_b = m$$

$$m_a \cdot a = m_b \cdot b$$

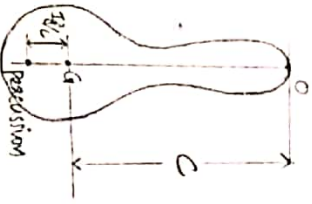
$$m_b = m \cdot \frac{a}{a+b}$$

$$I_G = \frac{m}{12} \left(\frac{l}{2} \right)^2$$

$$I_G = \frac{m}{12} \left(\frac{l}{2} \right)^2 = \frac{m}{12} \left(\frac{l}{2} \right)^2$$

$$I_G = m \cdot \frac{l^2}{12}$$

$$I_G = m \cdot \frac{l^2}{12}$$



$$I_G = m \cdot \frac{l^2}{12} + m \cdot c^2$$

$$I_G = m \cdot \frac{l^2}{12} + m \cdot c^2$$

rotation couple

$$= m \cdot (d-b) \cdot \omega$$

$$T_c = 10 (0.07 - 0.125) 385$$

$$= 5024 \text{ Nm}$$

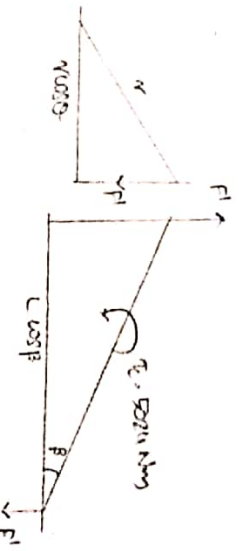
$$b = \frac{l}{2} = 0.125 \text{ m}$$

$$5024 = F' \cdot 0.0385$$

$$F' = 11219 \text{ N}$$

T_1 (Turning moment on bank due to the force of the superimposed mass)

$$= 5024 \text{ Nm}$$



T_2 (T.M available on bank due to F')

$$= F' \cdot 0.0385$$

$$= 814 \text{ Nm}$$

T_3 (T.M on the bank due to m on bank pin)

$$= m \cdot g \cdot 0.0385$$

$$= 41.2 \text{ Nm}$$



Net FM available on bank = 3739 Nm Q